Agentic AI — Study Notes

Notes on what Agentic AI is, how it differs from Generative AI, its key characteristics, and its core components.

1. What is Agentic AI?

Agentic AI is a type of AI that can take up a task/goal from a user and then work towards completing it on its own, with minimal human guidance. It plans, takes actions, adapts to changes, and seeks help only when necessary.

In simple terms: you give the system a goal, and the system itself figures out how to achieve that goal — handling both the planning and the execution with minimal human involvement. This is fundamentally different from other AI paradigms such as Generative AI chatbots (e.g., ChatGPT-style tools), which are reactive: at each step, the user asks a question and the chatbot answers exactly that question, with no initiative of its own.

Aspect	Generative AI Chatbot (Reactive)	Agentic AI System (Proactive)
Interaction style	Answers exactly what is asked, at the moment it is asked. No initiative of its own.	Given a high-level goal, then plans and executes all the sub-steps on its own.
Example (trip planning)	User must ask separately about dates, transport, hotels, itinerary — one question at a time.	User says “plan my Goa trip between these dates”; the system figures out transport, hotel recommendations, and a full itinerary on its own.
Monitoring	User must keep checking status manually and ask follow-up questions.	System proactively monitors progress and flags issues/decisions on its own (e.g., “not enough applicants, should I boost the post?”).

2. Example: Agentic AI for Hiring

Scenario: an HR recruiter is given the task of hiring a backend engineer, and has access to an Agentic AI chatbot to help with it.

Process:

- The recruiter specifies the goal and constraints: “Hire a backend engineer, remote, 2–4 years of experience.”
- The agent understands the goal and creates a plan: draft a job description (JD) → post it on job platforms → monitor applications → adjust if needed → screen candidates → schedule interviews → send offer letters → onboard selected candidates.
- The agent drafts the JD by accessing company documents (tech stack, salary bands, responsibilities) and asks for a quick review/approval.
- It posts the JD on chosen platforms (e.g., LinkedIn) using tool/API access, then proactively monitors applications.
- If applications are too few after a few days, it notices this on its own and proposes changes (e.g., broaden the role to “full-stack engineer”, run paid ads) — asking permission before spending money.
- Once enough applications arrive, it parses resumes with a resume-parser tool, shortlists strong candidates, and asks whether to proceed with interviews.
- It checks the recruiter's calendar, proposes interview slots, drafts and sends emails to both recruiter and candidates, sends reminders and a question bank.

- After the recruiter picks a candidate, it drafts an offer letter, sends it after review, monitors for acceptance, then sends a welcome email, submits IT/laptop provisioning requests, and offers to schedule an onboarding meeting.

Key takeaway: the most striking feature is autonomy — the human only supplies the goal (and occasional approvals); the agent does its own planning, execution, and adapts when something goes wrong.

3. Six Key Characteristics of Agentic AI

Any Agentic AI system can be identified/verified by checking whether it exhibits these six traits:

Trait	Meaning
Autonomy	Ability to make decisions and take actions on its own to achieve a goal, without step-by-step human instructions.
Goal-oriented	Operates with a persistent objective in mind and continuously directs its actions toward achieving it, rather than just responding to isolated prompts.
Planning	Ability to break down a high-level goal into a structured sequence of actions/sub-goals.
Reasoning	The cognitive process of interpreting information, drawing conclusions, and making decisions — needed during both planning and execution.
Adaptability	Ability to modify plans, strategies, and actions in response to unexpected conditions, while staying aligned with the goal.
Context awareness	Ability to understand, retain, and utilize relevant information from ongoing tasks, past interactions, user preferences, and environmental cues to make better decisions across a multi-step process.

3.1 Autonomy (in depth)

Autonomy means the AI system is proactive — it takes action before being asked, rather than only responding when prompted (reactive, like a typical chatbot). Autonomy shows up in multiple aspects of the system's operation:

- Execution autonomy — automatically executing each step of a plan one after another.
- Decision-making autonomy — e.g., deciding on its own how many/which candidates to shortlist.
- Tool-usage autonomy — deciding on its own which tool to use for a given step.

Because full autonomy can be risky (e.g., sending offers with incorrect salaries, biased shortlisting, or spending money without permission), autonomy is usually controlled via:

- Scope of permissions — limiting which tools/actions the agent can perform independently.
- Human-in-the-loop checkpoints — requiring human approval before continuing at certain risky steps.
- Override controls — letting users stop, pause, or change the agent's behavior at any time.
- Guardrails and policies — hard rules and ethical boundaries the agent must always follow (e.g., “never schedule interviews on weekends”, “never use informal language”).

3.2 Goal-Oriented (in depth)

The goal acts like a compass that directs the system's autonomy — without a goal, autonomous behavior has no direction. Goals can be:

- Independent — e.g., “hire a backend engineer.”
- Constrained — e.g., “hire remotely only”, “must be based in India”, “budget capped at $X.”

Goals (with their constraints, status, progress, and creation time) are typically stored in the agent's core memory, often in a structured format (e.g., JSON), and can be altered mid-way — for example, switching from “hire a full-time backend engineer” to “find a freelancer” if the original approach isn't working, which then changes the downstream planning and execution.

3.3 Planning (in depth)

Agentic AI systems operate in two broad stages: planning (deciding how to achieve a goal) and execution (carrying out that plan). This two-step cycle is iterative — if a step turns out to be infeasible mid-way, the system goes back to planning, creates a revised plan, and continues executing, repeating until the goal is achieved.

Planning itself happens in three steps:

1. Generate multiple candidate plans — planning is treated like a search problem: given an initial state and a desired final state, there are usually multiple possible paths, so the system lays out more than one candidate plan (e.g., “post on job portals” vs. “use internal referrals / a hiring agency”).
2. Evaluate the candidate plans against criteria such as: efficiency (which is faster), tool availability (is the required tool/API even available), cost, risk of failure, and alignment with constraints (e.g., which option yields more remote candidates).
3. Select the best plan — either via human-in-the-loop input (asking a human supervisor to pick) or via a pre-programmed policy that decides automatically.

3.4 Reasoning (in depth)

Reasoning is the cognitive process through which the system interprets information from its environment, draws conclusions, and makes decisions — needed in both the planning stage and the execution stage.

Reasoning is required during planning for:

- Goal decomposition — breaking a task into a series of executable steps.
- Tool selection — figuring out which external tool is needed for a given step (e.g., needing a search tool to look up market salary data).
- Resource/time/dependency/risk estimation for the plan.

Reasoning is required during execution for:

- Decision-making when multiple options are available (e.g., interview 2 or 3 shortlisted candidates?).
- Deciding when to act independently vs. when to involve a human (human-in-the-loop judgment calls).
- Error handling — e.g., if a tool/API is down, deciding whether to retry later, notify a human, or use an alternative platform.

3.5 Adaptability (in depth)

Adaptability is the agent's ability to modify its plans, strategies, and actions in response to unexpected conditions, while staying aligned with the original goal. Common triggers for needing adaptability:

- Tool/API failures — e.g., calendar API is down, so the agent asks the human directly for availability instead.
- External feedback from the environment — e.g., very few applicants after a few days, prompting a change in approach (ads, revised JD).
- Mid-way changes to the goal itself — e.g., switching from hiring a full-time engineer to hiring a freelancer.

3.6 Context Awareness (in depth)

Context awareness is the agent's ability to understand, retain, and use relevant information from ongoing tasks, past interactions, user preferences, and environmental cues throughout a multi-step (often multi-day) process. Without this, a long-running process (like hiring, which can span many days) simply cannot function correctly.

Typical things kept in context include: the original goal, progress so far and past conversation with the human supervisor, the current state of the environment, responses from tools (e.g., “candidate B has 3 years of Django + AWS experience”), user-specific preferences, and any policies/guardrails in effect.

This is implemented via memory, which is generally of two types:

- **Short-term memory:** information relevant to the current session (e.g., a tool's response that a specific candidate has certain experience).
- **Long-term memory:** information that persists across sessions (e.g., a guardrail like “never send an offer letter without approval”).

4. Core Components of an Agentic AI System

Most Agentic AI applications are built from five high-level components (there can be more, but these are the most fundamental):

Component	Role
Brain (LLM)	Interprets the goal, performs planning (breaking the goal into sub-goals), performs reasoning (in both planning and execution), decides which tool to use, and handles natural-language communication with the human. Does most of the “heavy lifting.”
Orchestrator	Executes the plan step by step: task sequencing, conditional routing between steps, retry logic on tool failure, looping/iteration, and delegation (deciding when a task goes to the LLM vs. to a human). Acts like the “nervous system” / project manager of the application.
Tools	How the agent interacts with the external world — API calls, database changes, sending emails, retrieving documents/knowledge bases (RAG). Act like the agent's “hands and legs.”
Memory	Short-term memory holds current-session information (messages, tool calls, immediate decisions). Long-term memory holds high-level goals, past interactions, user preferences, and cross-session decisions. Also handles state tracking (how much progress has been made).
Supervisor	Implements human-in-the-loop: requests human approval before high-risk actions, enforces guardrails, and escalates edge cases to a human for review.

Note: each of these can be broken down further — for example, the “Brain” component often internally contains a planner (that generates multiple candidate plans) and an evaluator (that scores/selects among them).

5. Quick Revision Summary

4. Agentic AI takes a goal from a user and works towards completing it autonomously, with minimal human guidance — unlike reactive Generative AI chatbots that only respond to exact prompts.
5. Six defining characteristics: autonomy, goal-oriented behavior, planning, reasoning, adaptability, and context awareness. If a system exhibits all six, it qualifies as Agentic AI.
6. Autonomy needs to be controlled via permission scoping, human-in-the-loop checkpoints, override controls, and guardrails/policies, since unchecked autonomy can be risky.
7. Planning is a 3-step process: generate multiple candidate plans → evaluate them (efficiency, tool availability, cost, risk, constraint alignment) → select the best one.
8. Reasoning underlies both planning (goal decomposition, tool selection, estimation) and execution (decision-making, human-in-the-loop judgment, error handling).
9. Five core components: Brain (LLM) for interpretation/planning/reasoning, Orchestrator for step-by-step execution and control flow, Tools for interacting with the external world, Memory (short-term and long-term) for context and state tracking, and Supervisor for human-in-the-loop approvals, guardrails, and escalations.